First assignment

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This is the first assignment for MAT242. As the course materials, announcements, etc. are in English, I will be writing this in English. That is my preference anyways. Hopefully the use of the word ”write-up” doesn’t necessitate this being super-duper structured and dull, but that I am free to share my exploratory thought process. Still I will strive to be as rigorous as possible, of course.

Note: I am submitting one day late as I thought the deadline was the 7th. My mistake, I have been away at a conference and things got a bit hectic.

The problem

If I understood the problem correctly, I am to show that the sort of ”canonical” way of defining a topology on the cartesian product of some sets X_i is the only way of doing it when we want to satisfy that universal property, whatever that means!

I have heard enough waffling on Category Theory to know that this term is relevant there and I was immediately confused by the ”there is only one way to do it” line. A cursory google search on ”product topology vs. box topology” immediately brought up how this so-called ”universal property” fails to be satisfied by the box topology when looking at the cartesian product of infinitely many X_i.

The universal property

This property is stated for the cartesian product of two sets so I will begin there. I will denote the projection pr$_i$ by $\pi_i : X_1 \times X_2 \to X_i$ because it looks cooler and writing \operatorname a bunch is tedious.

Let X_1, X_2 be sets. The cartesian product has the property that for any function $f : Z \to X_1 \times X_2$, f is uniquely defined by the two projections

$$f_1 := \pi_1 f : Z \to X_1 \text{ and } f_2 := \pi_2 f : Z \to X_2,$$

where $\pi_i : X_1 \times X_2 \to X_i$ is the projection $(x_1, x_2) \overset{\pi_i}{\mapsto} x_i$ for $i = 1, 2$.

This essentially says something quite trivial which is that "to specify where z goes in $X_1 \times X_2$, specify its X_1-coordinate and its X_2-coordinate".

We would want the same to hold for topological spaces. If X_1, X_2, Z are topological spaces, then a function $f : Z \to X_1 \times X_2$ is continuous if and only if f_1 and f_2 are.

There is probably some nice way to formulate the above as requiring the some diagram commutes or something, but I don't know Category Theory as of right now.

What we now want to do is to show that this requirement forces a unique topology on $X_1 \times X_2$ called the product topology.

The product topology

So we are essentially given that this is the product topology. That means that what we should try to do now is to prove that for arbitrary spaces X_1, X_2, Z, the product topology has the desired property.

Definition (Product topology). *Let X_1 and X_2 (for notational consistency) be topological spaces, meaning there exist some $\tau_1 \subseteq \mathcal{P}(X_1)$ and $\tau_2 \subseteq \mathcal{P}(X_2)$ such that (X_1, τ_1) and (X_2, τ_2) are topological spaces.*
Consider

$$\mathcal{B} = \{U \times V \mid U \subseteq X_1 \text{ open}, \text{ and } V \subseteq X_2 \text{ open}\}.$$

The product topology on $X_1 \times X_2$ is the topology having $\mathcal{B}$ as a basis.
So an arbitrary open set of $X_1 \times X_2$ is a union of $U \times V$:

$$W = \bigcup_{\alpha \in A} U_\alpha \times V_\alpha.$$

Munkres also gives an equivalent definition using projections, very exciting.

Now suppose are given some continuous function $f : Z \to X_1 \times X_2$, meaning that for every open set $W \in X_1 \times X_2$ we have that

$$f^{-1}(W) = \{z \in Z \mid f(z) \in W\}$$

is open in Z.

Now we should think back to what this means under the universal property of the cartesian product, since this must necessarily hold for f. Let $f_i := \pi_i f : Z \times X_i$, $(i = 1, 2)$. Then $f(z) = (f_1(z), f_2(z))$. The idea now becomes quite clear. We assume f is continuous, we know the projections are continuous (though we should probably prove so) and continuity is preserved under composition (maybe we ought to prove that aswell or maybe it won't be necessary) so f continuous $\Rightarrow f_1, f_2$ continuous. Then we must prove the converse which isn't that bad either.

> **Lemma.** *Let X_1, X_2, Z be topological spaces and endow $X_1 \times X_2$ with the product topology. If*
>
> $$f : Z \to X_1 \times X_2$$
>
> *is continuous, then*
>
> $$f_1 = \pi_1 \circ f, \ \ f_2 = \pi_2 \circ f$$
>
> *are continuous.*

Proof. Suppose $f : Z \to X_1 \times X_2$ is continous. Then, for any open set in $X_1 \times X_2$,

$$f^{-1}(W) = \{z \in Z \mid f(z) \in W\}$$

is open in Z.

Take f_1. Suppose $U \subseteq X_1$ is open. Then

$$\begin{aligned} f_1^{-1}(U) &= (\pi_1 \circ f)^{-1}(U) \\ &= f^{-1}(\pi_1^{-1}(U)). \end{aligned}$$

Then since

$$\pi_1^{-1}(U) = U \times X_2,$$

and U is open in X_1 by assumption, and X_2 is open in itself, we have that $U \times X_2$ is open in the product topology.

Therefore since f is continuous, we have that

$$f_1^{-1}(U) = f^{-1}(U \times X_2)$$

is open in Z.

The proof of continuity for f_2 is effectively identical. □

Perhaps it would have been good enough to remark that f_i is the composition of continuous functions, but who knows.

> **Lemma.** *Let X_1, X_2, Z be topological spaces and again give $X_1 \times X_2$ the product topology. If*
>
> $$f_1 = \pi_1 \circ f, \ \ f_2 = \pi_2 \circ f$$
>
> *are continuous then*
>
> $$f(z) = (f_1(z), f_2(z))$$
>
> *is continuous.*

Proof. Suppose $W \subseteq X_1 \times X_2$ is open.

We want to show that

$$f^{-1}(W)$$

is open.

Recall that

$$f^{-1}(W) = f^{-1}\left(\bigcup_{\alpha \in A} U_\alpha \times V_\alpha\right),$$

where U_α, V_α are open in X_1, X_2, respectively, and for all α.

Recall also that

$$\begin{aligned} f^{-1}\left(\bigcup_\alpha U_\alpha \times V_\alpha\right) &= \bigcup_\alpha f^{-1}(U_\alpha \times V_\alpha) \\ &= \bigcup_\alpha f_1^{-1}(U_\alpha) \cap f_2^{-1}(V_\alpha), \end{aligned}$$

since

$$\begin{aligned} z \in f^{-1}(U_\alpha \times V_\alpha) &\iff f(z) \in U_\alpha \times V_\alpha \\ &\iff (f_1(z), f_2(z)) \in U_\alpha \times V_\alpha \\ &\iff f_1(z) \in U_\alpha \text{ and } f_2(z) \in V_\alpha \\ &\iff z \in f_1^{-1}(U_\alpha) \cap f_2^{-1}(V_\alpha). \end{aligned}$$

Thus the preimage of an open set in $X_1 \times X_2$ is the union of preimages (under f_1, f_2) of open sets in X_1 and X_2. Since f_1, f_2 are continuous by assumption, $f^{-1}(W)$ is a union of open sets in Z and therefore open. □

Theorem. *For topological spaces X_1, X_2, Z where we once again endow $X_1 \times X_2$ with the product topology,*

$$f : Z \to X_1 \times X_2$$

is continuous if and only if

$$\pi_1 \circ f, \;\; \pi_2 \circ f$$

are continuous.

Proof. Apply the previous two lemma. □

If Category Theory has a more compact way of showing that the product topology indeed has this property I shall work tirelessly against those who label it as ”abstract nonesense”.

It has to be the product topology!

Again we are still working with simple cartesian products of only two topological spaces.

Now what should the intuition be here, I wonder? Say we have some spaces X_1, X_2 and some topology on $X_1 \times X_2$. Why must it be that for any Z and any functions $f : Z \to X_1 \times X_2$, f cont. iff f_1, f_2 cont. must necessarily imply that $X_1 \times X_2$ has the product topology?

Well, this condition must hold for every Z and f. So let $Z = X_1 \times X_2$ with the product topology (which we now knows satisfies the nice property itself), and let $f = \mathrm{id}$.

From here we can get a simple argument!

Theorem. *Let τ be a topology on $X_1 \times X_2$. Suppose that for every topological space Z and every $f : Z \to X_1 \times X_2$, f is continuous iff $\pi_1 f, \pi_2 f$ are continuous. Then $\tau = \tau_{\mathrm{prod}}$.*

Proof. Let τ_{prod} denote the product topology on $X_1 \times X_2$.

We first show that

$$\tau \subseteq \tau_{\mathrm{prod}}.$$

Consider

$$Z = (X_1 \times X_2, \tau_{\mathrm{prod}})$$

and the identity map

$$\mathrm{id} : (X_1 \times X_2, \tau_{\mathrm{prod}}) \to (X_1 \times X_2, \tau).$$

Its coordinate maps are

$$\pi_1 \circ \mathrm{id} = \pi_1, \qquad \pi_2 \circ \mathrm{id} = \pi_2.$$

The projections π_1 and π_2 are continuous with respect to the product topology. Hence, by the assumed property of τ, the identity map is continuous.

Therefore every τ-open set is τ_{prod}-open, and so

$$\tau \subseteq \tau_{\mathrm{prod}}.$$

We now prove the reverse inclusion. Consider

$$Z = (X_1 \times X_2, \tau)$$

and the identity map

$$\mathrm{id} : (X_1 \times X_2, \tau) \to (X_1 \times X_2, \tau_{\mathrm{prod}}).$$

By applying the assumed property of τ to the identity map

$$\mathrm{id} : (X_1 \times X_2, \tau) \to (X_1 \times X_2, \tau),$$

we see that the projections

$$\pi_1 : (X_1 \times X_2, \tau) \to X_1, \qquad \pi_2 : (X_1 \times X_2, \tau) \to X_2$$

are continuous.

Since the product topology satisfies the universal property, continuity of these two coordinate maps implies that

$$\mathtt{id} : (X_1 \times X_2, \tau) \to (X_1 \times X_2, \tau_{\mathtt{prod}})$$

is continuous. Hence

$$\tau_{\mathtt{prod}} \subseteq \tau.$$

Therefore

$$\tau = \tau_{\mathtt{prod}}.$$

□

Very cool!

After this I was thinking back to the box topology thing which, to be honest, I haven't looked into that much. I was curious though to what I found about the box topology not satisfying this property when the cartesian product is infinite. After a quick search I found out that this topology is apparently the same as the product topology for finite cartesian product so that turned out to be relatively uninteresting.